

Rajiv V. Joshi

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Education

Carnegie Mellon University	Pittsburgh, PA
Master of Science in Mechanical Engineering - Research GPA: 4.0/4.0	May 2026
Virginia Tech	Blacksburg, VA
Bachelors of Science in Mechanical Engineering - Robotics and Mechatronics GPA: 3.64/4.0	May 2024
Bachelors of Science in Mechanical Engineering - Mechanical GPA: 3.64/4.0	May 2024

Coursework

Masters: ADV Mechatronics (TA) | Haptic Design Interfaces | Modern Control Theory | Computer Vision for Engineers (In-Progress) | ML and AI for Engineers (In-Progress)

Undergraduate: Robotics and Automation | Kinematics | Differential Equations | Deforms | Dynamics

Skills

Embedded Systems: Arduino | STM 32 | Jetson Orin Nano | Oscilloscope | DMM | Serial Communication (SPI, I2C, CAN)

Manufacturing: PCB Design | Soldering | Circuit Schematics | Rapid Prototyping | 3-D printing | Machining

Software/Programming: Python | C/C++ | MATLAB | SolidWorks | Fusion 360 | KiCAD | ANSYS | Linux Command Line

General: Hardware/Software Debugging, Firmware programming, Fast-paced environments, Cross-team communication

Research Experience

MetaMobility Lab at Carnegie Mellon University	Pittsburgh, PA
<i>Knee Exoskeleton – MS Research Project</i>	April 2025 - Present

- Coordinated and led a team of 5 students in research, design, and manufacturing a lightweight knee exoskeleton capable of delivering 20 Nm of torque directly to knee joints for people suffering from osteoarthritis or gait impairment
- Utilized a Teensy 4.1 to implement SPI protocol, in C/C++, with IMU sensors through CAT6 connectors and twisted pair cables to shield signal from noise, increasing gait cycle estimator loop frequency from 100Hz to 500Hz
- Debugged multiple hardware and firmware faults using an Oscilloscope and DMM to visualize logic signals, voltage values, and circuit continuity to rapidly diagnose causes and iterate/manufacture immediate solutions

MetaMobility Lab at Carnegie Mellon University	Pittsburgh, PA
<i>Hip Exoskeleton – MS Research</i>	May 2024 - Present

- Optimized hip exoskeleton hardware for improved fitting to users referencing human biomechanics concepts, SolidWorks, and 3D printing, yielding comfortable motor-joint alignment and 25% reduced system weight
- Redesigned circuit schematics and PCB designs to integrate embedded systems such as MCUs, IMUs, and FSRs via KiCAD, reducing connections from 8 to 4 while maintaining robust connections and signal integrity
- Integrated CAN bus and I2C protocols, in Python and C/C++, with motors encoders and IMUs, enabling gait cycle estimation for controllers and hardware torque control upwards of 18 Nm, cutting user hip work by 20% to 30%
- Communicated with various global manufacturers for pricing and outsourcing manufacturing of Carbon Fiber and Nylon based parts designed in SolidWorks considering basic DFM and DFA concepts

Academic Projects

Carnegie Mellon University	Pittsburgh, PA
<i>Haptic Design Interfaces: 2 DoF Haptic Hand Interface – Course Project</i>	April 2025 - May 2025

- Developed a open-source compact haptic fingers device, with a team of 6 graduate students, meshing STM32 MCU, 4 low torque motors, and 3D-printed parts as a teaching tool for robot kinematics, biological sensors, and haptics
- Visualized a finger joint workspace in MATLAB to define mechanical operational ranges of a capstan-driven end-effector, controlled in C/C++ by applying PID and forward kinematics to render virtual 3D objects

Carnegie Mellon University	Pittsburgh, PA
<i>ADV Mechatronics: Embedded System Game Design – Course Project</i>	November 2024 - December 2024

- Implemented PWM, timer interrupts, and event driven state-machine on an STM32 for controlling a custom H-bridge motor driver circuit, in whack-a-mole style arcade game designed with a team of 3
- Lead creation of detailed circuit schematics, using KiCAD, for guiding protoboard soldering/manufacturing, optimizing cable routing between components, and assisting debugging circuit/logic issues with an oscilloscope and DMM